
*This talk contains examples that may be offensive

Contrast Comes at a Cost: Prompting Large Language Models for Perception-Centered Gendered Style Transfer

Hongyu Chen, Neele Falk, Michael Roth, Agnieszka Faleńska

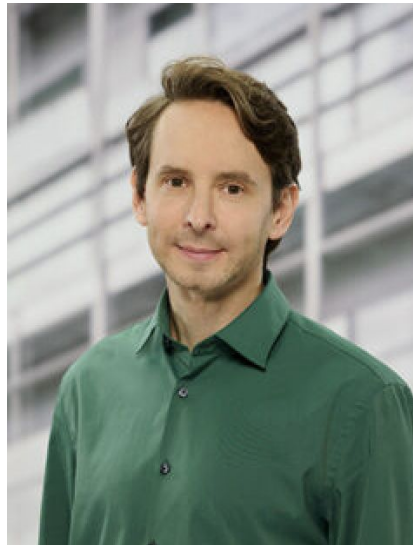
Interchange Forum for Reflecting on Intelligent Systems, University of Stuttgart
Institute for Natural Language Processing (IMS), University of Stuttgart
Natural Language Understanding Lab, University of Technology Nuremberg

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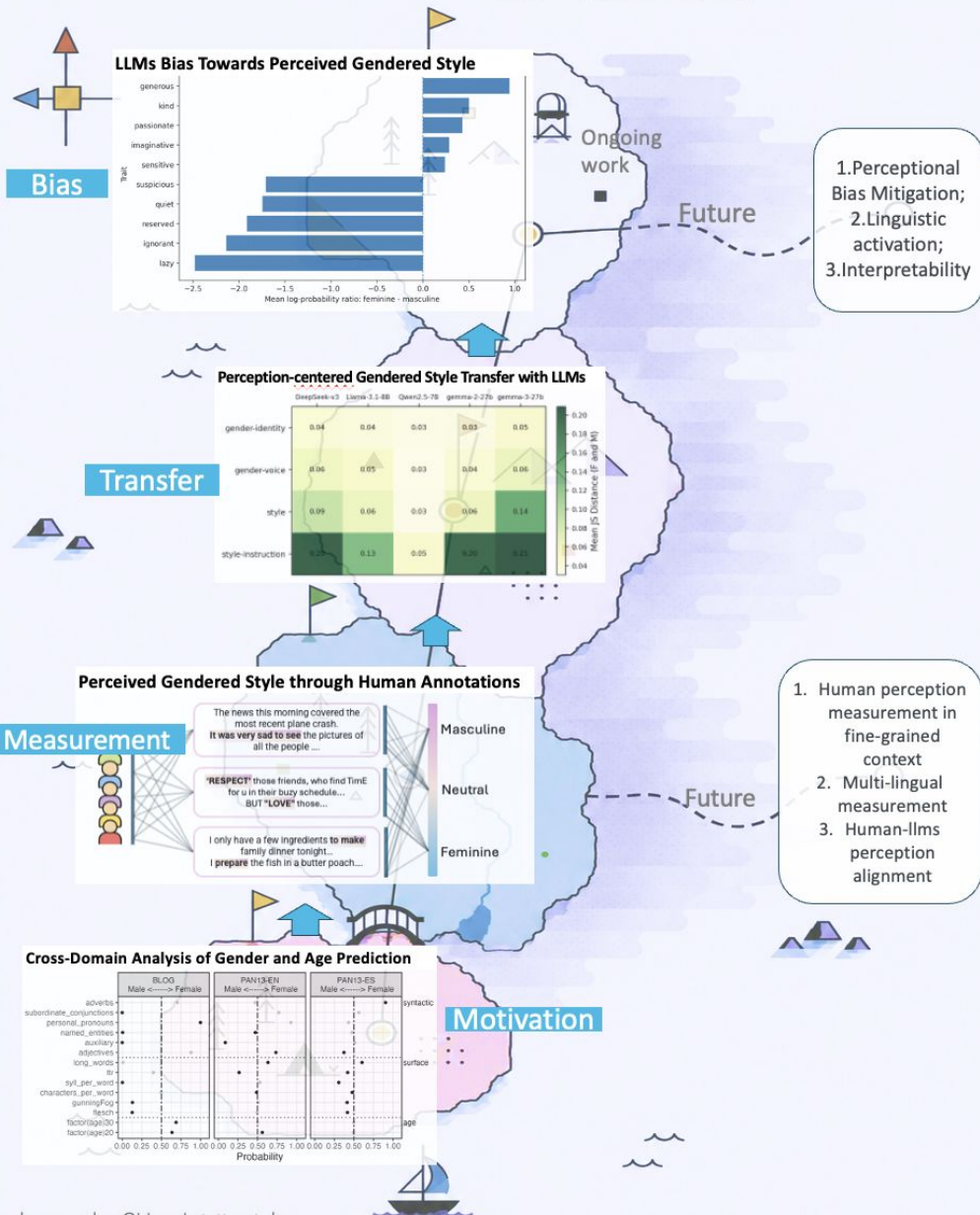
Universität Stuttgart

Hongyu Chen

Supervisor: Dr. Anieszka Falańska

Prof. Dr. Michael Roth

Perceived Gendered Style: Measurement, Transfer and Bias with LLMs



Chen et al, 2025

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Text style transfer

In a linguistically grounded view, **style** refers to the manner in which semantics is expressed (McDonald and Pustejovsky, 1985).

Text style transfer is a controllable text generation task that changes how a text is expressed while preserving its meaning (Jin et al., 2022).

Text style transfer

Offensive Non-offensive
what a **f**kng** circus this is $\rightleftarrows$ what a **big** circus this is

Text style transfer

Offensive
what a **f**kng** circus this is

Non-offensive
what a **big** circus this is



Impolite
[] send me the data []

Polite
could you **please** send me the data



Text style transfer

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what a **f**kng** circus this is $\rightleftarrows$ what a **big** circus this is

Impolite Polite
 send me the data $\rightleftarrows$ **could** you **please** send me the data

Informal Formal
The kid is **freaking out**. $\rightleftarrows$ The child is **distressed**

Text style transfer – Applications

Writing support

Writing Assistant

You

Please rewrite this email to sound warmer and more affectionate while preserving its meaning:

Hi Alex, thank you for visiting yesterday. I enjoyed spending time with you. I hope we can meet again soon.

Assistant

Hi Alex,

It was lovely to see you yesterday—thank you for coming over! I really enjoyed our time together, and I'm looking forward to seeing you again soon.

Message...

Media Language Style Correction - Neutral

ABIGAIL BEALL SCIENCE OCT 2, 2018 9:55 AM

The Nobel prize for physics has been awarded to a woman for the first time in 55 years



Donna Strickland wins Nobel Prize in Physics for laser research, becoming the first woman laureate in 55 years.

“I really enjoyed the cozy atmosphere of this café”



“The café has a pleasant atmosphere”

“I really enjoyed the cozy atmosphere of this café”



Gendered text style transfer

“The café has a pleasant atmosphere”

Female author

“I really enjoyed the cozy atmosphere of this café”



Gendered text style transfer

Male author

“The café has a pleasant atmosphere”

-
- It is often framed as converting between male- and female-authored texts (Kang et al., 2019; Sudhakar et al., 2019).

Female author

“I really enjoyed the cozy atmosphere of this café”



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Female Male
"I really enjoyed the cozy atmosphere of this café" $\rightleftarrows$ "The café has a pleasant atmosphere"

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Female Male
"I really enjoyed the cozy atmosphere of this café" $\rightleftarrows$ "The café has a pleasant atmosphere" subjective, less cut-off indicators

Text style transfer

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what a **big** circus this is

Impolite
[] send me the data []

Polite
could you **please** send me the data

Relatively clear style indicator

Informal
The kid is **freaking out**.

Formal
The child is **distressed**

Female
“I really enjoyed the cozy atmosphere of this café”

Male
“The café has a pleasant atmosphere”

subjective, less cut-off indicators

Gendered style transfer task

- It is often framed as converting between male- and female-authored texts (Kang et al., 2019; Sudhakar et al., 2019).

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“I really enjoyed the cozy atmosphere of this café”



Gendered text style transfer

Male author

“The café has a pleasant atmosphere!”

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Gendered text style transfer

Male author

“The café has a pleasant atmosphere!”

- **author identity** and **gender expression** are distinct constructs

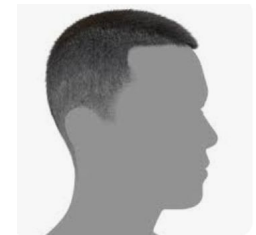
-
- **Gender identity:**
 - individual's internal sense and how they identify themselves



-
- **Gender expression:**
 - how individuals present their gender externally (e.g., feminine, masculine, or gender-neutral)



men can wear eyeliner



women can have buzz cuts

≠ gender identity

Gendered style transfer task

- It is often framed as converting between male- and female-authored texts (Kang et al., 2019; Sudhakar et al., 2019).

Female author

“I really enjoyed the cozy atmosphere of this café”

≠ Perceived feminine



Gendered text style transfer

Male author

“The café has a pleasant atmosphere!”

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How do we know the transfer between gendered styles actually happens

- author identity and gender expression are distinct constructs
- transferring between male-authored and female-authored texts does not necessarily correspond to transferring between masculine and feminine expression

Methods and Evaluation on Gendered Style Transfer in Previous Work:

Reddy and Knight (2016)

- Replaced words to obscure the author's gender and defeat a gender classifier.

Sudhakar, Upadhyay, and Maheswaran (2019)

- Used a three-stage **Delete–Retrieve–Generate** pipeline
 - deleted style-bearing words using attention from a BERT-based classifier;
 - retrieved target-style attributes from sentences with similar content;
 - generated the final sentence with a GPT-based decoder-only transformer.

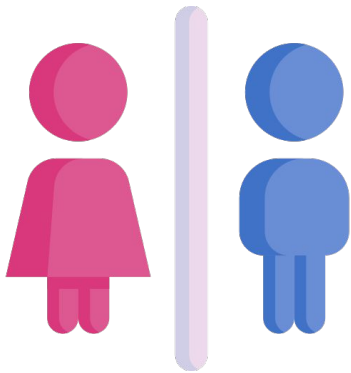
Kang, Gangal, and Hovy (2019) — PASTEL

- Challenged the assumption that content and style can be cleanly separated.
- Found that style classifiers rely on both stylistic and content-related features.
- Transferred multiple persona attributes jointly, such as gender and education.

Style-transfer success: typically measured using the proportion of outputs assigned to the target class by a pretrained classifier.

Human evaluation: generally assesses fluency and meaning preservation; perceived gender transfer is often omitted because annotators may find it difficult or unreliable to judge.

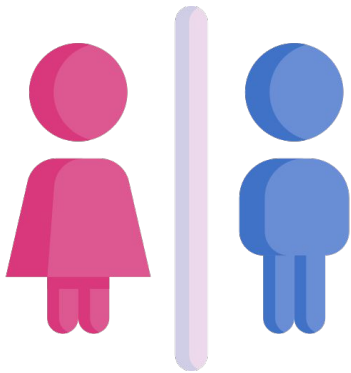
reduce gendered style transfer to a binary classification problem



Overlook the spectrum of gender expression and style variation from readers' interpretation



reduce gendered style transfer to a binary classification problem

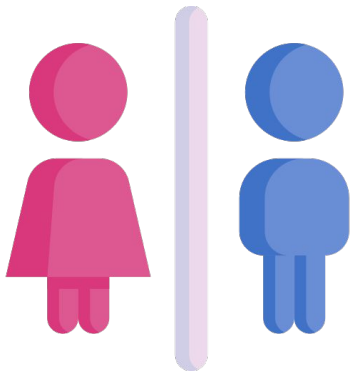


Overlook the spectrum of gender expression and style variation from readers' interpretation



Eg. a male author's writing can also be perceived as feminine style

reduce gendered style transfer to a binary classification problem



Overlook the spectrum of gender expression and style variation from readers' interpretation



Eg. a male author's writing can also be perceived as feminine style

Then what should “gendered style” mean in a style-transfer task?

So what do we do?

Following Chen et al. (2025), we introduce the notion of **perceived gendered style**: how readers interpret the gendered attributes of a text rather than relying on author information.

- given an input text t and a target style $s \in \{\textit{feminine}, \textit{masculine}\}$, the system produces a rewrite t' that preserves the intended meaning of t while shifting linguistic realization toward s .

- LLM-based Style Transfer - controlled text generation

Prior work includes personal or authorial style, formality, sentiment, and detoxification transfer, but not yet explored with perceived gendered style transfer

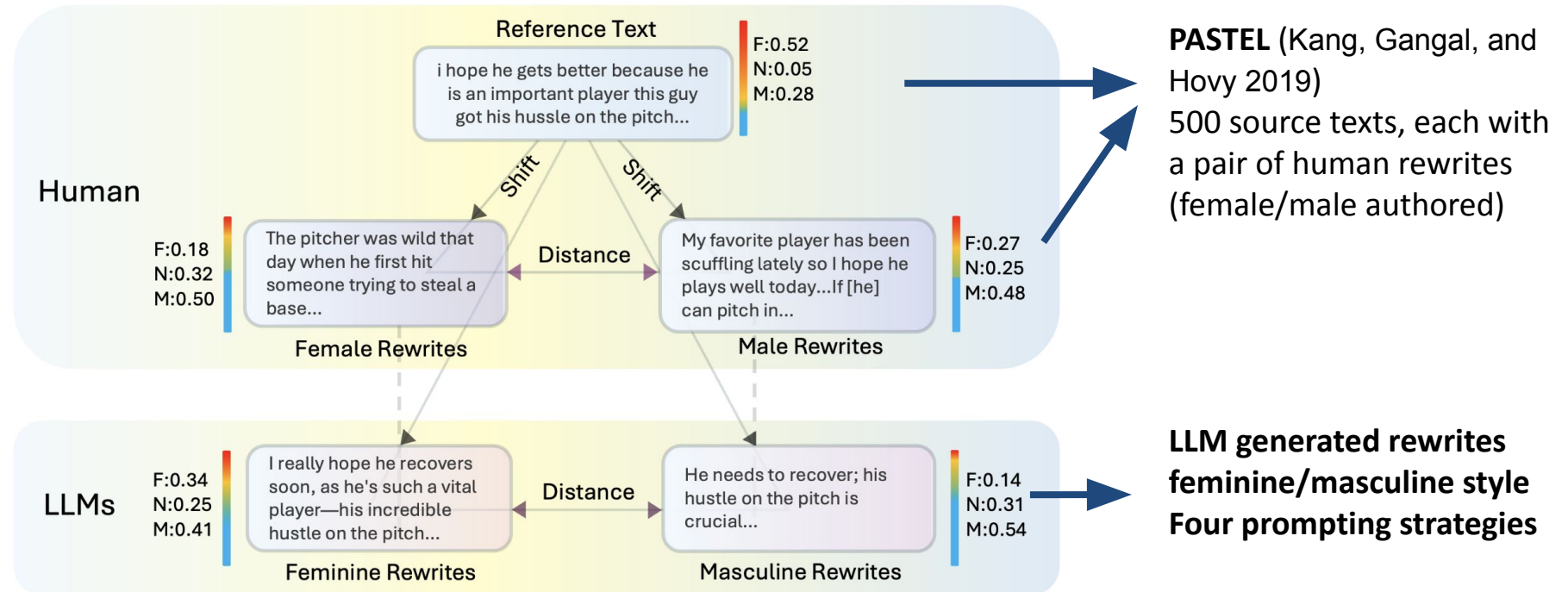
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Dataset



- each reference text t is paired with two rewrites targeting feminine and masculine style
- rewrites come either from humans with self-reported gender or from LLMs.

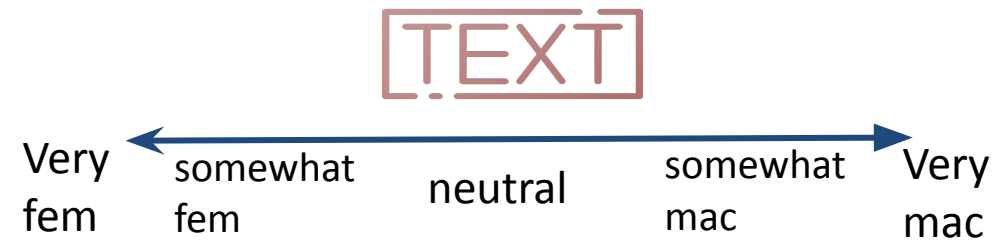
Gendered Style Operationalization

- 1 Author identity:  An author who identifies as a woman or a man (Morales Sánchez et al., 2022)
- 2 Gendered voice:  Female-author or male-author voice tendencies: stance (Saito, 2010)
- 3 Gendered style:  Feminine or masculine linguistic tendencies (Troiano et al., 2023)
- 4 Fine-Grained Style: * instruction based on average feminine/masculine style explicit cues such as hedging, directness, stance and relational framing (chen et al, 2025)

All conditions preserve meaning, prohibit new information and stereotypes, and keep the rewrites comparable in length.

Evaluation of Perceived Gendered Style

perceived gendered-style classifier trained on the dataset of Chen et al. (2025)



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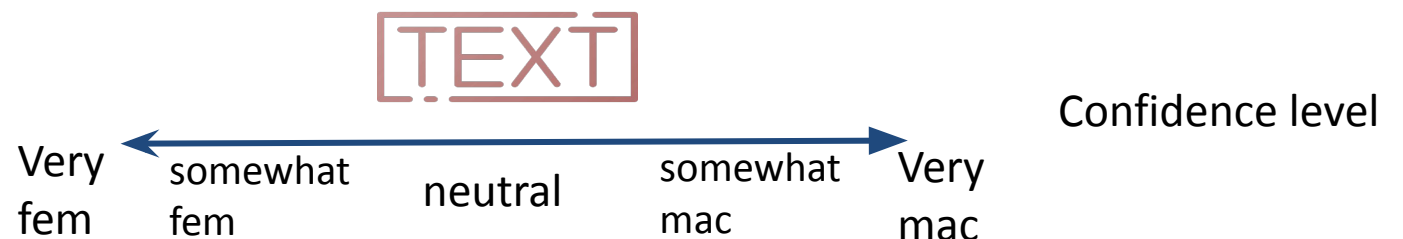


We collapse the style scale into **three** categories:

F (very/somewhat feminine), N (neutral), and M (very/somewhat masculine),

Evaluation of Perceived Gendered Style

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We collapse the style scale into **three** categories:

F (very/somewhat feminine), N (neutral), and M (very/somewhat masculine),

and train a confidence-weighted BERT-based classifier (Devlin et al., 2018) to predict a three-way probability distribution for a text t :

$$p(\cdot \mid t) = [p(F \mid t), p(M \mid t), p(N \mid t)]$$

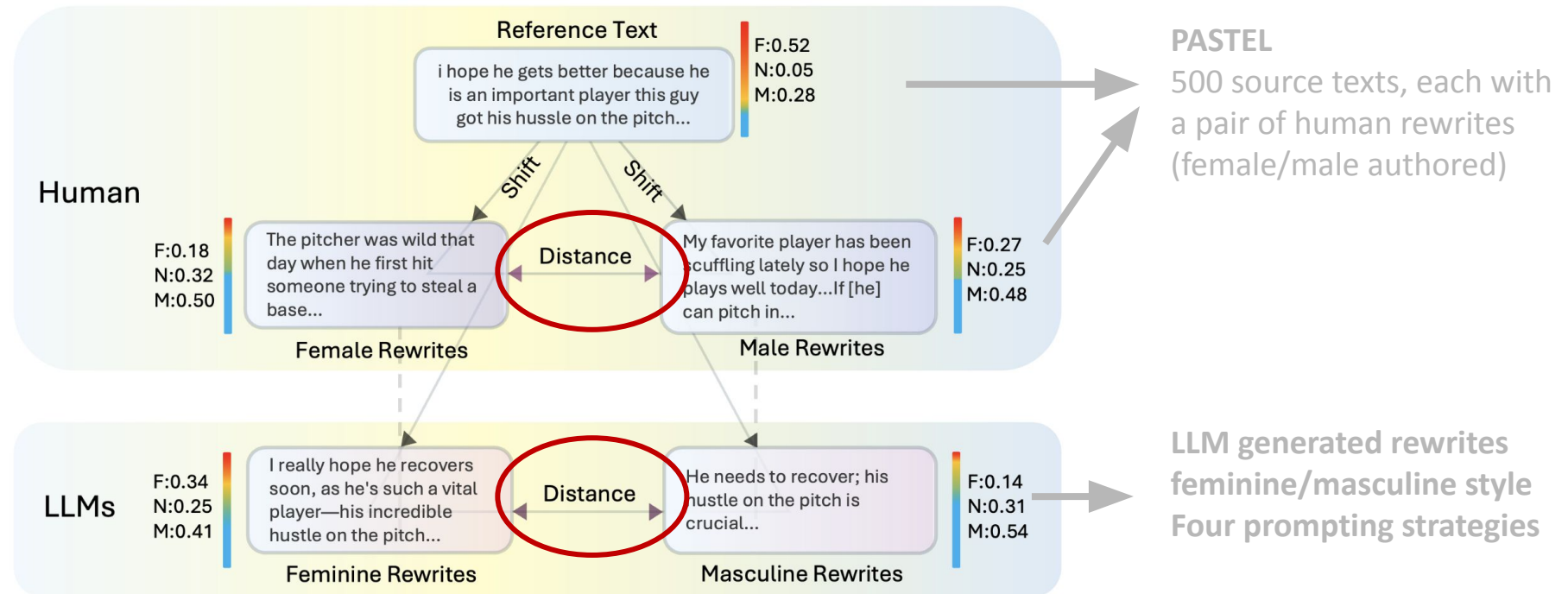
Model	Mean JS Distance	Std. JS Distance
Random Forest	0.279	0.135
MLP (key_extracted_features)	0.253	0.133
BERT-base-uncased	0.205	0.105
BERT-base-uncased (conf.-weighted)	0.204	0.104

Comparison of models (soft labels) by mean Jensen–Shannon (JS) distance and standard deviation of JS distance. Lower is better.

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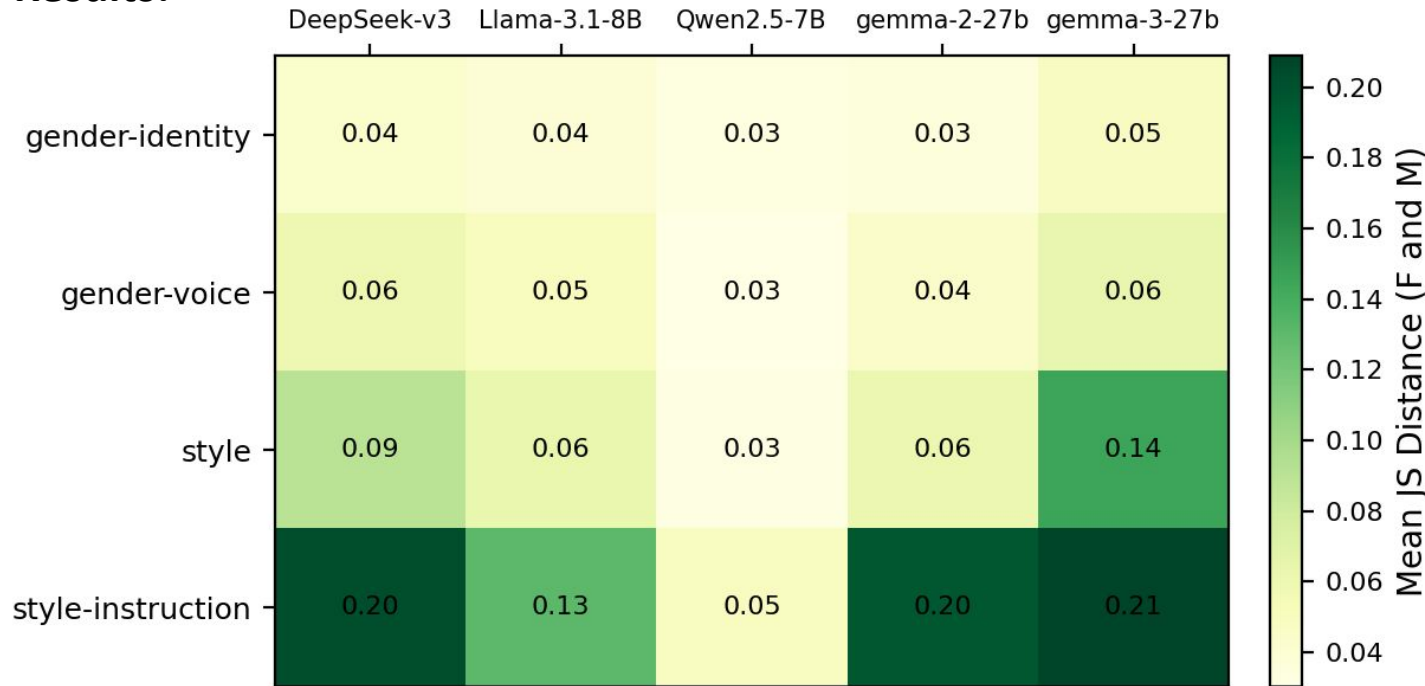
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Style Distance

Given a paired rewrite (t'_F, t'_M) for the same reference t , we quantify how strongly the two rewrites contrast in perceived gendered style using Jensen-Shannon distance

$$D_{\text{style}}(t'_F, t'_M) = \text{JS}(p(\cdot | t'_F) || p(\cdot | t'_M))$$

Results:



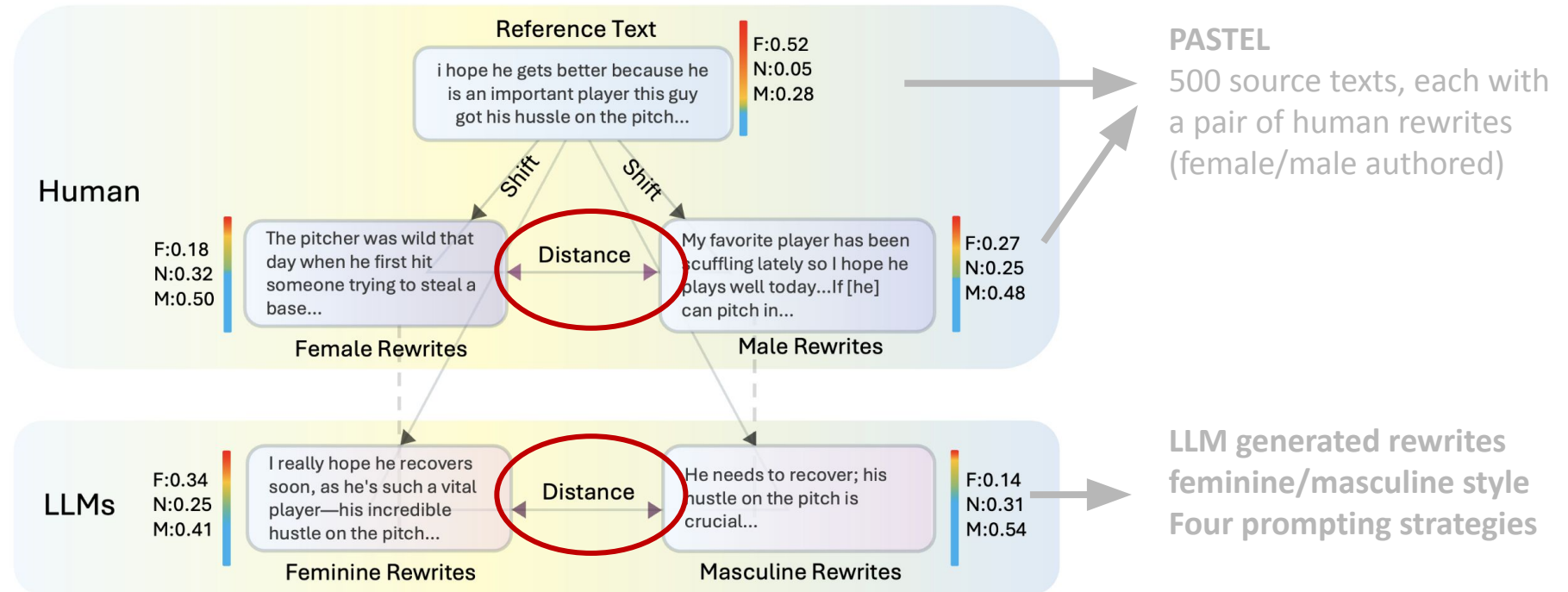
Style Distance between perceived feminine and masculine style.

The average style distance for human rewrite pairs is .12.

Identity prompting generally produces the weakest contrast.

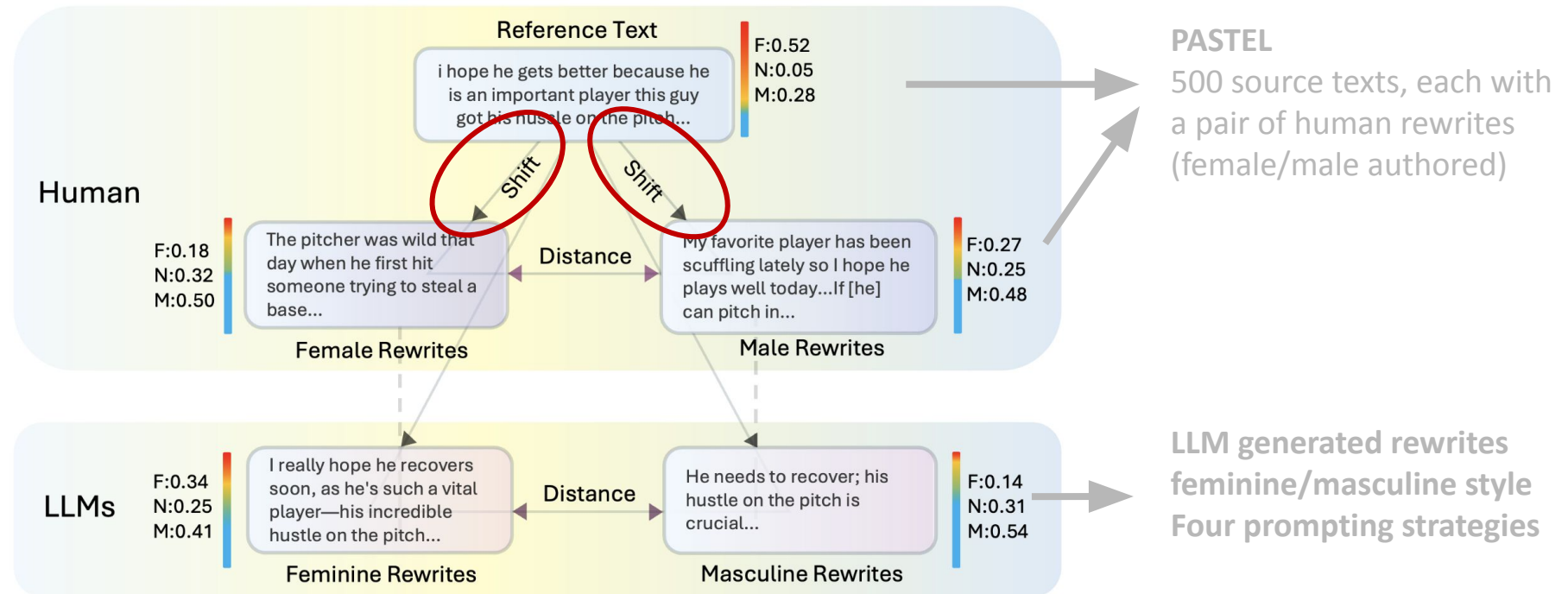
Fine-grained instructions outperform general style prompts (t-test) and human rewrite for most models.

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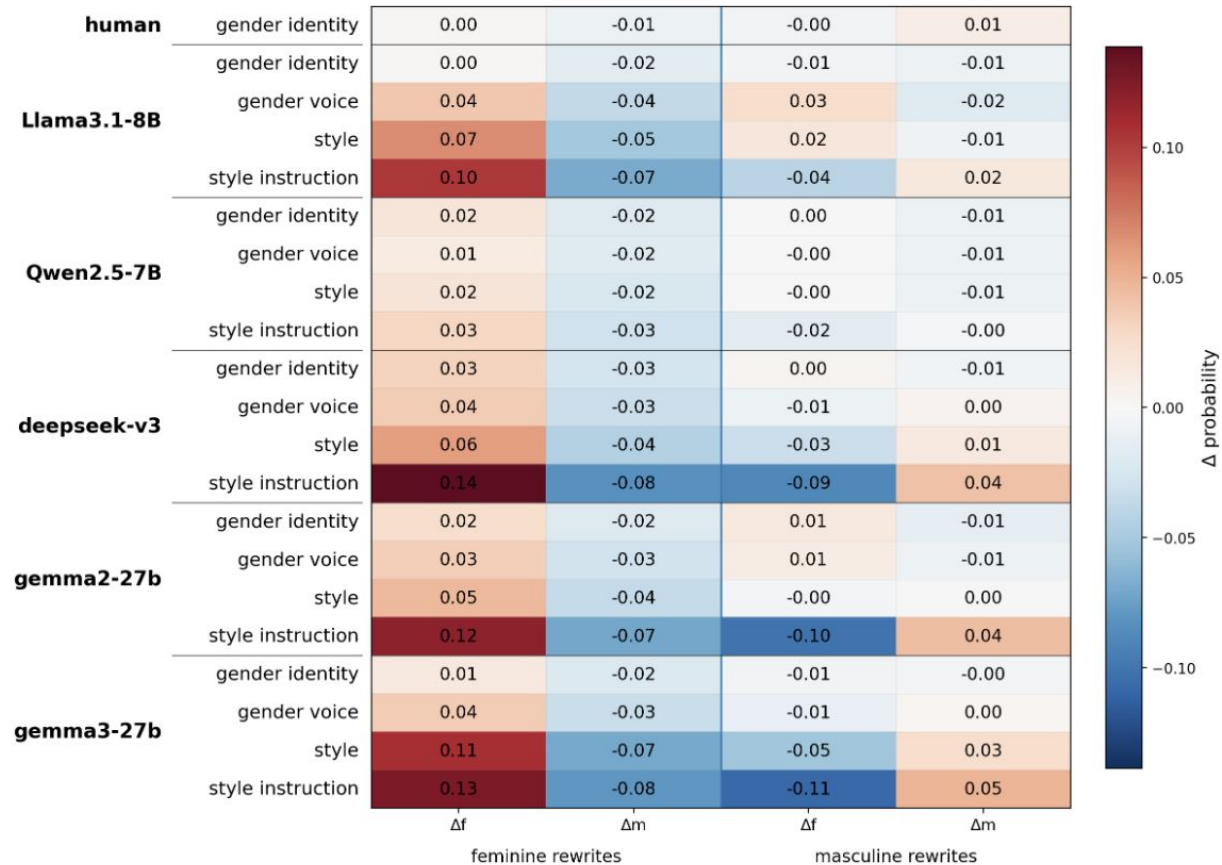
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Style Shift

Results:



- Feminine-targeted shifts are generally stronger than masculine-targeted shift.
- Observed strong pairwise contrast does not imply equal shift in both directions.

Style Shift relative to reference text. Each cell reports $\Delta s = p(s | t') - p(s | t)$ for style class $s \in \{F, M\}$. Red denotes a positive shift (higher probability than the reference), and blue denotes a negative shift.

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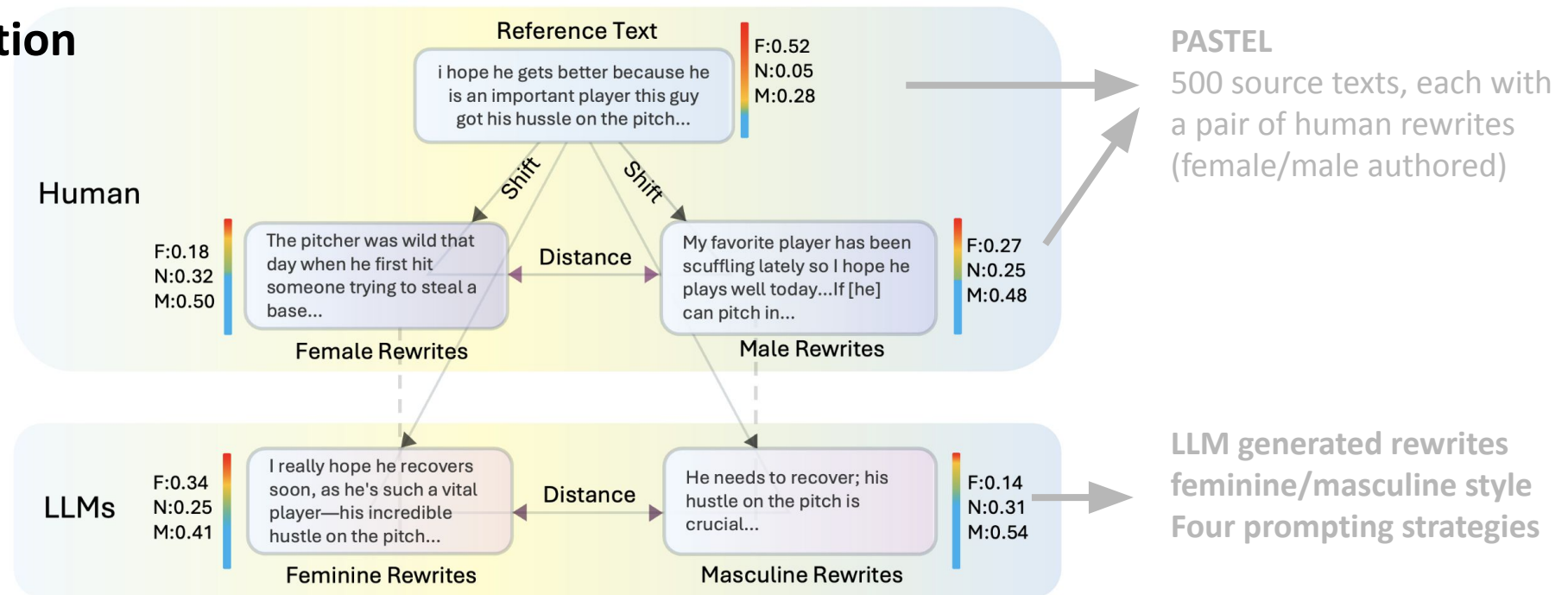
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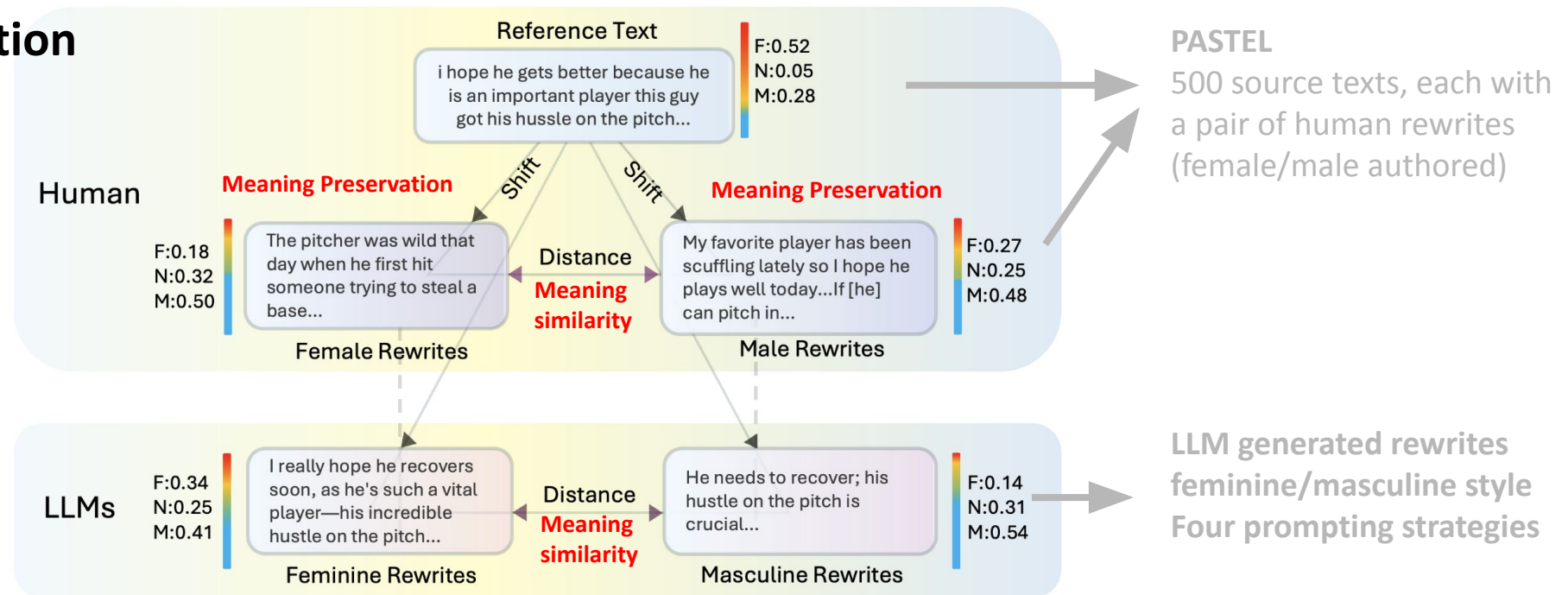
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Meaning preservation



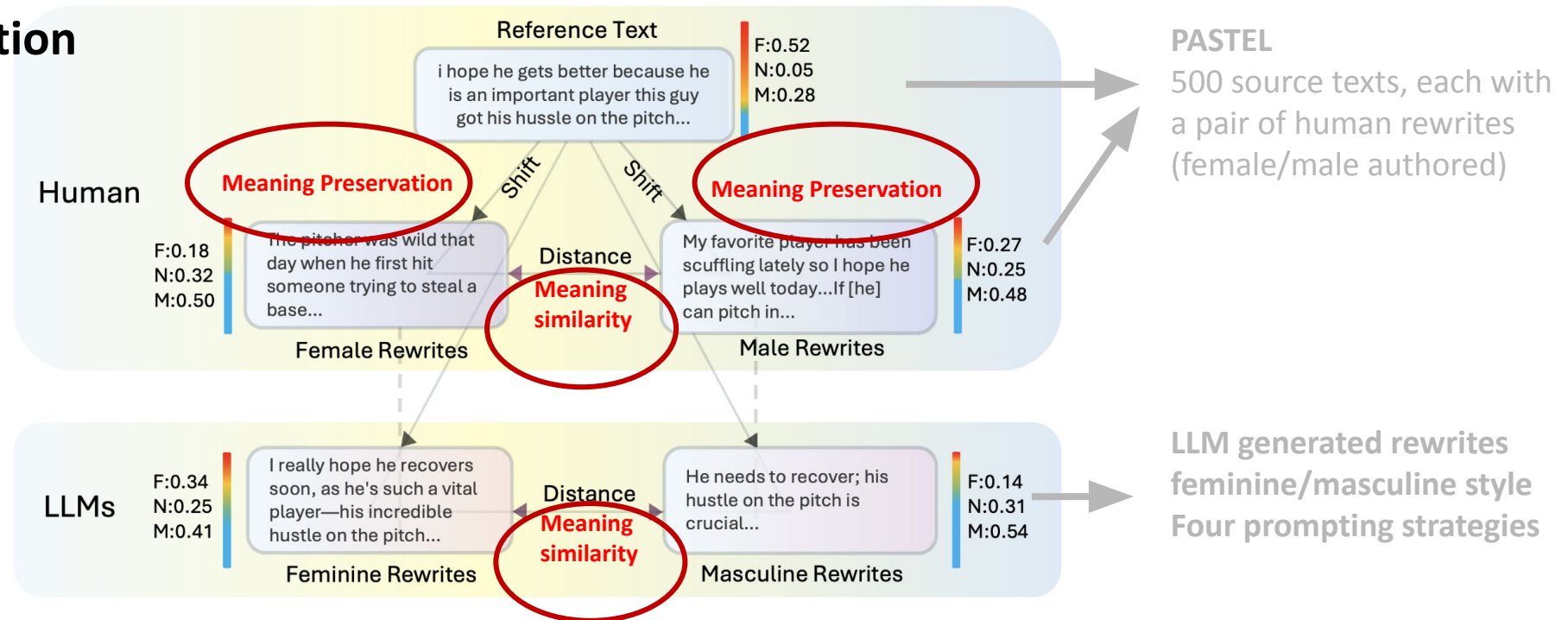
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Meaning Similarity and Preservation

we use natural language inference model WANLI (Liu et al., 2022) to estimate entailment probabilities for

(i) meaning similarity within the rewrite pair (t'_F , t'_M) and

(ii) Meaning preservation of each rewrite relative to the reference t . For a pair of texts (t'_F , t'_M), we report a bidirectional entailment score defined as the minimum of the two directional entailment probabilities.

$$e(t'_F, t'_M) = p_{\text{NLI}}(\text{entail} \mid t'_F \rightarrow t'_M)$$
$$\text{EntailMin}(t'_F, t'_M) = \min(e(t'_F, t'_M), e(t'_M, t'_F))$$

Results:

Approach	Meaning Similarity (masc↔fem)	Meaning Preservation (masc→ref)	Meaning Preservation (fem→ref)
human	0.052 ± 0.18	0.060 ± 0.14	0.053 ± 0.13
LLM_identity	0.802 ± 0.33	0.780 ± 0.33	0.776 ± 0.34
LLM_voice	0.779 ± 0.34	0.752 ± 0.34	0.754 ± 0.34
LLM_style	0.712 ± 0.37	0.742 ± 0.34	0.715 ± 0.36
LLM_style_instruction	0.585 ± 0.40	0.739 ± 0.35	0.585 ± 0.39

Average meaning similarity (WANLI NLI) within feminine-masculine rewrite pairs and meaning preservation of each rewrite relative to the reference text.

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➤ Human produce lowest meaning similarity/preservation

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A trade-off between style contrast and semantic fidelity

Example:

Target	Text	Metrics
Reference	i just left work when i came upon a car that flipped over . it was so close to the cement blockade but it didn't quite reach it . there was no one in the car thank goodness but it looked pretty beat up . there were other people around besides me to see if the owners needed help . i left the scene shortly after taking pictures .	PGS: [0.328, 0.400 ,0.272] Grammar: 0.803
Style instruction		
Feminine	I had just finished work when I came across a car that had flipped over. It was so incredibly close to the cement barrier, but thankfully didn't quite hit it. To my relief , there was no one inside, though the car looked quite damaged. A few other caring people were also nearby, checking if the owners needed assistance. I took some pictures before leaving the scene shortly after.	PGS: [0.399 , 0.310, 0.291] MP: 0.438 MS: 0.772 Grammar: 0.925
Masculine	After work , I encountered an overturned car near a cement blockade. It narrowly missed the barrier. No one was in the vehicle, but it was heavily damaged. Others were present to offer help. I documented it with photos and then left.	PGS: [0.221, 0.428 , 0.351] MP: 0.687 MS: 0.772 Grammar: 0.972
Style		
Feminine	I had just finished work when I encountered a flipped-over car. It was positioned so near the concrete barrier, though it hadn't actually collided with it. Thankfully , no one was inside the vehicle, but it appeared quite damaged. A few other individuals were also present, checking if the owners required assistance. I departed the area not long after capturing some photos.	PGS: [0.314, 0.362 , 0.324] MP: 0.748 MS: 0.973 Grammar: 0.824
Masculine	I'd just left work when I saw a car that had flipped over. It was really close to the cement barrier but didn't hit it. Nobody was in the car, luckily , though it looked pretty banged up . Other people were around too, seeing if the owners needed help. I left shortly after taking pictures.	PGS: [0.364 , 0.347, 0.289] MP: 0.932 MS: 0.973 Grammar: 0.966

PGS= [p(F), p(M), p(N)] Color highlights mark lexical changes in generated rewrites:

olive frequent paraphrases in references; **teal** content from reference paraphrased in that rewrite; **blue** new phrases not present in the reference.

Human Evaluation - Do Human confirm the trade-off?

Survey 1: Style Contrast

Pair 1 of 10

Text A

Our group always rents bikes whenever we vacation. We explored so much territory and captured countless photos. Numerous ships sailed past this island harbor. This region isn't reachable by car. We paused here for a break since we had journeyed many miles. Back at the hotel, we snapped some pictures while chatting about dinner options.

Text B

Our group rents bikes on every vacation. We covered extensive ground and took heaps of photos. Many ships passed by the island port. This area is inaccessible by car. We stopped here to rest after traveling numerous miles. At the hotel, we took photos while discussing dinner plans.

Style contrast

How strongly contrasted are these two texts in feminine vs masculine style?

1: Not contrasted at all 2: Slightly contrasted 3: Moderately contrasted 4: Strongly contrasted 5: Very strongly contrasted

No value selected yet.

Meaning similarity

To what extent do the two texts express the same meaning or content?

1: Completely different 2: Mostly different 3: Partly similar 4: Mostly same 5: Same meaning

Prolific, each pair has 4 participants, 50 pairs/ style_condition and fine_grained_style (deepseek v3)

Survey 2 : Perceived Gendered Style

[Text 1] I just had the most wonderful time at the Halloween party last night! It felt so lively with so many people joining in, and some of the costumes were absolutely hilarious and fun. I loved capturing all those special moments with my camera, and I really hope we can plan something just as magical again next year.

Select a scale:

1: Very Feminine 2: Somewhat Feminine 3: Neutral 4: Somewhat Masculine 5: Very Masculine

No value selected yet.

Confidence Level

Choose an option

Comments (Optional)

Back

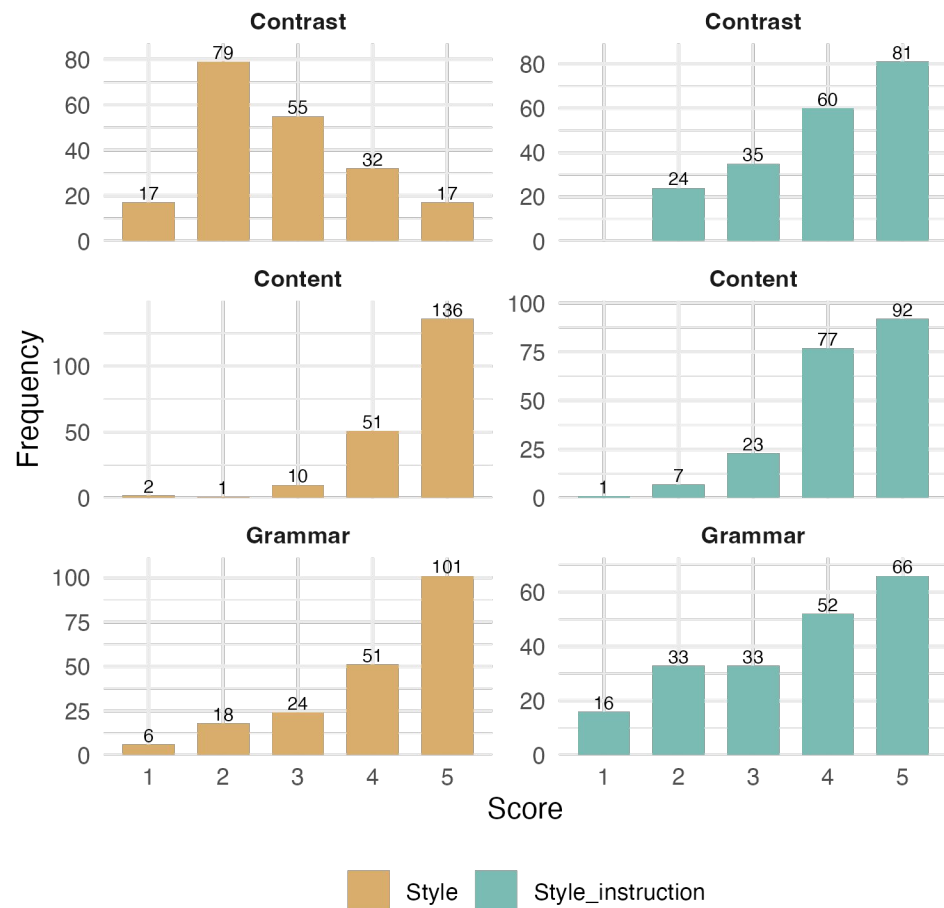
Next

Completed 0 out of 40 regular texts.

Prolific, each individual text has 4 participants, 200 texts from style_condition and fine_grained_style (deepseek v3)

Human Evaluation - Do Human confirm the trade-off?

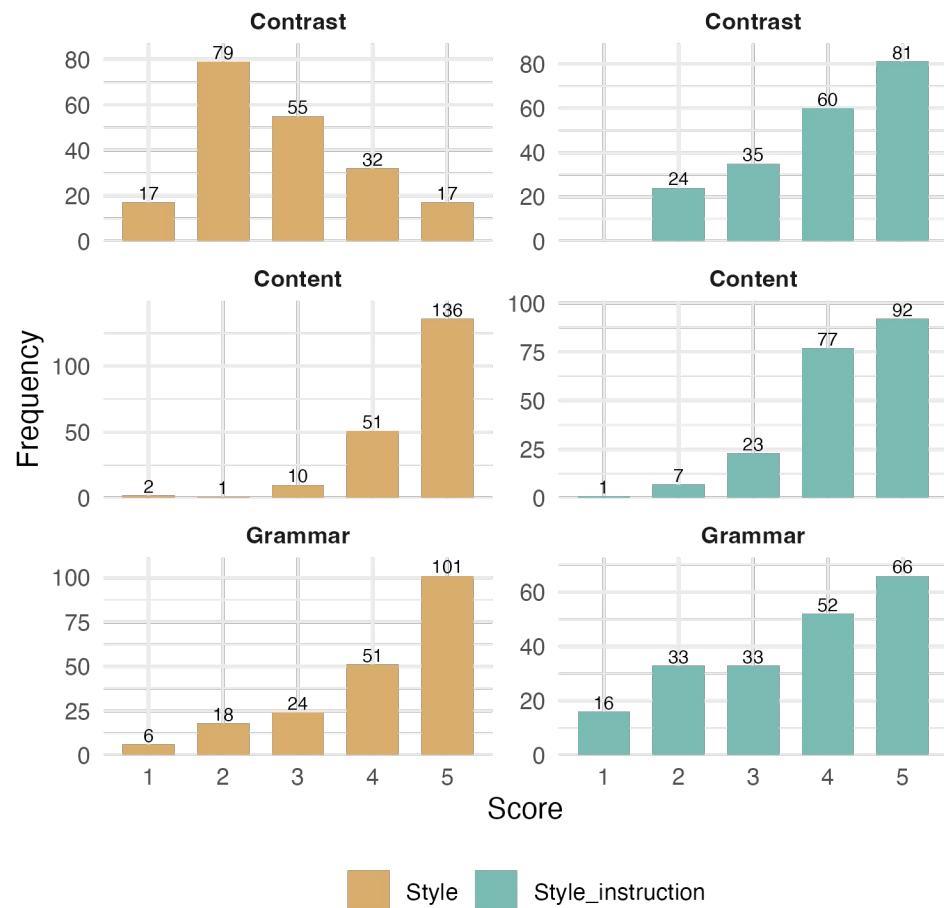
Results:



Frequency Distribution of human annotation scores for stylistic contrastiveness, meaning similarity (Content), and grammaticality across the style and style instruction conditions from DeepSeek-V3 (50 pairs each condition).

Human Evaluation - Do Human confirm the trade-off?

Results:

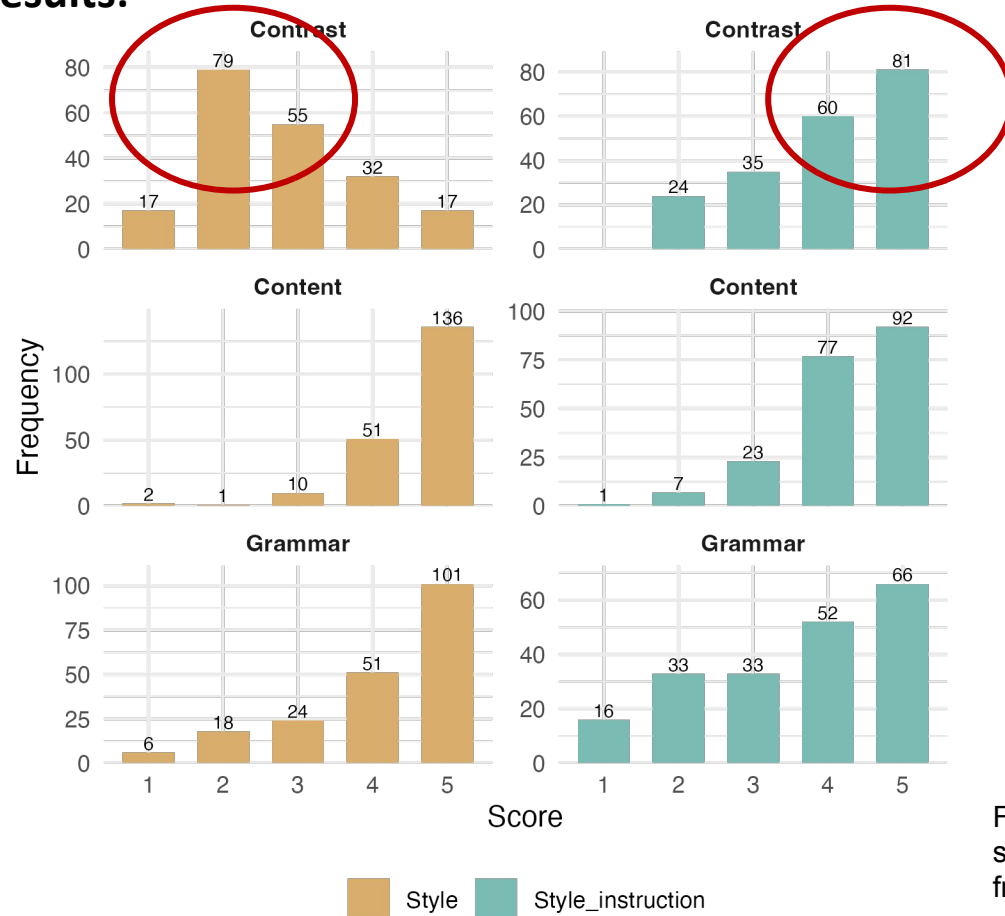


➤ Higher contrast scores indicate stronger perceived gendered stylistic differences between feminine and masculine rewrites, **Average contrast score (3.99 vs. 2.77)**

Frequency Distribution of human annotation scores for stylistic contrastiveness, meaning similarity (Content), and grammaticality across the style and style instruction conditions from DeepSeek-V3 (50 pairs each condition).

Human Evaluation - Do Human confirm the trade-off?

Results:

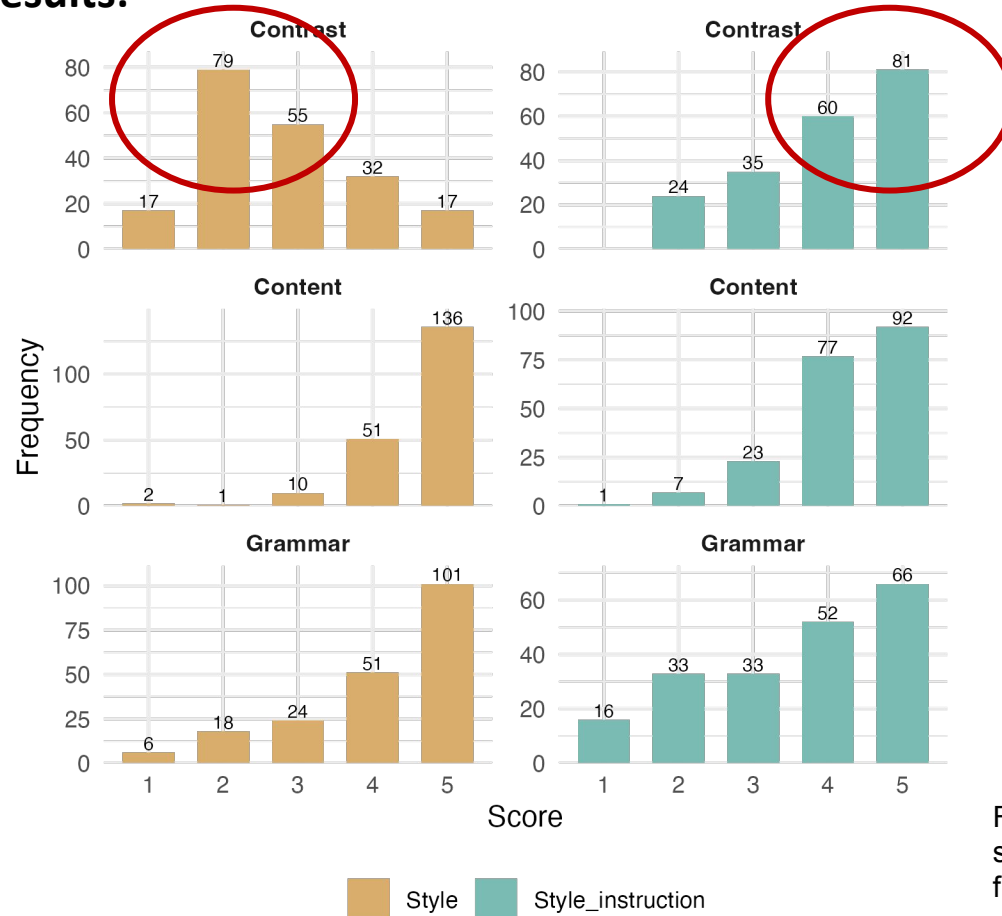


➤ Higher contrast scores indicate stronger perceived gendered stylistic differences between feminine and masculine rewrites, **Average contrast score (3.99 vs. 2.77)**

Frequency Distribution of human annotation scores for stylistic contrastiveness, meaning similarity (Content), and grammaticality across the style and style instruction conditions from DeepSeek-V3 (50 pairs each condition).

Human Evaluation - Do Human confirm the trade-off?

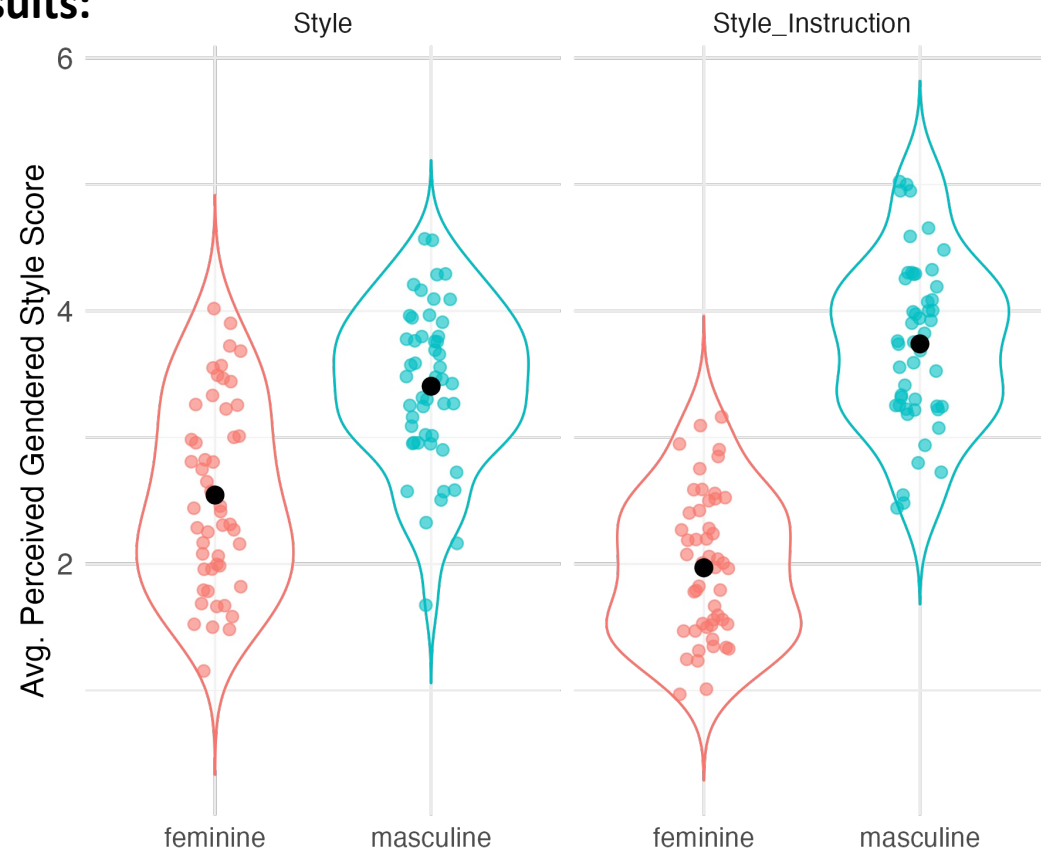
Results:



- Higher contrast scores indicate stronger perceived gendered stylistic differences between feminine and masculine rewrites, **Average contrast score (3.99 vs. 2.77)**
- higher content and grammar scores indicate better semantic preservation and grammatical acceptability **Average meaning similarity score: (4.26 vs. 4.59)**

Frequency Distribution of human annotation scores for stylistic contrastiveness, meaning similarity (Content), and grammaticality across the style and style instruction conditions from DeepSeek-V3 (50 pairs each condition).

Results:



Each dot represents the average perceived gendered style score assigned by four annotators of one individual text.

Human annotations of average perceived gendered style scores for feminine and masculine rewrites under the style and style instruction conditions generated by DeepSeek-V3 (50 pairs each condition).

Takeaway

- LLMs can perform perceived style transfer
- But prompt operationalization matters
- Fine-grained style instructions yield high contrast but kill semantic fidelity